

Mode 6 — Square

DP32 (8×8) → DP16 (8×8) → DP16 (8×4) → DP8 (8×4) → DP8 (4×4) → add-and-square

Step 1 — DP32 (8×8)

$$\text{DP32 } (8 \times 8) = \sum_{i=0}^{31} a_i b_i$$

Step 2 — DP16 (8×8)

$$\text{DP32 } (8 \times 8) = \underbrace{\sum_{i=0}^{15} a_i b_i}_{\text{DP16 } (8 \times 8)} + \underbrace{\sum_{i=16}^{31} a_i b_i}_{\text{DP16 } (8 \times 8)}$$

Step 3 — DP16 (8×4)

$$\text{DP32 } (8 \times 8) = 2^4 \underbrace{\sum_{i=0}^{15} a_i b_i^{hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=0}^{15} a_i b_i^{lo}}_{\text{DP16 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=16}^{31} a_i b_i^{hi}}_{\text{DP16 } (8 \times 4)} + \underbrace{\sum_{i=16}^{31} a_i b_i^{lo}}_{\text{DP16 } (8 \times 4)}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP32 } (8 \times 8) = & 2^4 \underbrace{\sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=8}^{15} a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_i b_i^{hi}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=16}^{23} a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)} + \underbrace{\sum_{i=24}^{31} a_i b_i^{lo}}_{\text{DP8 } (8 \times 4)} \end{aligned}$$

Step 5 — DP8 (4×4)

$$\begin{aligned} \text{DP32 } (8 \times 8) = & 2^8 \underbrace{\sum_{i=0}^7 a_{H,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=0}^7 a_{L,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=8}^{15} a_{H,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{L,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} \\ & + 2^4 \underbrace{\sum_{i=0}^7 a_{H,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=0}^7 a_{L,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=8}^{15} a_{H,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=8}^{15} a_{L,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} \\ & + 2^8 \underbrace{\sum_{i=16}^{23} a_{H,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=16}^{23} a_{L,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^8 \underbrace{\sum_{i=24}^{31} a_{H,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{L,i} b_i^{hi}}_{\text{DP8 } (4 \times 4)} \\ & + 2^4 \underbrace{\sum_{i=16}^{23} a_{H,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=16}^{23} a_{L,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + 2^4 \underbrace{\sum_{i=24}^{31} a_{H,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} + \underbrace{\sum_{i=24}^{31} a_{L,i} b_i^{lo}}_{\text{DP8 } (4 \times 4)} \end{aligned}$$

Step 6 — Add-and-square (substitute each DP8 (4×4) = $\frac{1}{2} [\sum (x_i + b_i)^2 - \sum x_i^2 - \sum b_i^2]$)

$$\text{DP32 } (8 \times 8) = 2^8 \underbrace{\frac{1}{2} \left(\sum_{i=0}^7 (a_{H,i} + b_i^{hi})^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 } (4 \times 4)}$$

$$\begin{aligned}
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i} + b_i^{hi})^2 - \sum_{i=0}^7 a_{L,i}^2 - \sum_{i=0}^7 (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i} + b_i^{hi})^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i} + b_i^{hi})^2 - \sum_{i=8}^{15} a_{L,i}^2 - \sum_{i=8}^{15} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{H,i} + b_i^{lo})^2 - \sum_{i=0}^7 a_{H,i}^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=0}^7 (a_{L,i} + b_i^{lo})^2 - \sum_{i=0}^7 a_{L,i}^2 - \sum_{i=0}^7 (b_i^{lo})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{H,i} + b_i^{lo})^2 - \sum_{i=8}^{15} a_{H,i}^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + \frac{1}{2} \underbrace{\left(\sum_{i=8}^{15} (a_{L,i} + b_i^{lo})^2 - \sum_{i=8}^{15} a_{L,i}^2 - \sum_{i=8}^{15} (b_i^{lo})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=16}^{23} (a_{H,i} + b_i^{hi})^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=16}^{23} (a_{L,i} + b_i^{hi})^2 - \sum_{i=16}^{23} a_{L,i}^2 - \sum_{i=16}^{23} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^8 \frac{1}{2} \underbrace{\left(\sum_{i=24}^{31} (a_{H,i} + b_i^{hi})^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=24}^{31} (a_{L,i} + b_i^{hi})^2 - \sum_{i=24}^{31} a_{L,i}^2 - \sum_{i=24}^{31} (b_i^{hi})^2 \right)}_{\text{DP8 (4}\times\text{4)}} \\
& + 2^4 \frac{1}{2} \underbrace{\left(\sum_{i=16}^{23} (a_{H,i} + b_i^{lo})^2 - \sum_{i=16}^{23} a_{H,i}^2 - \sum_{i=16}^{23} (b_i^{lo})^2 \right)}_{\text{DP8 (4}\times\text{4)}}
\end{aligned}$$

$$\begin{aligned}
& \underbrace{+ \frac{1}{2} \left(\sum_{i=16}^{23} (a_{L,i} + b_i^{lo})^2 - \sum_{i=16}^{23} a_{L,i}^2 - \sum_{i=16}^{23} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + 2^4 \underbrace{\frac{1}{2} \left(\sum_{i=24}^{31} (a_{H,i} + b_i^{lo})^2 - \sum_{i=24}^{31} a_{H,i}^2 - \sum_{i=24}^{31} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}} \\
& + \underbrace{\frac{1}{2} \left(\sum_{i=24}^{31} (a_{L,i} + b_i^{lo})^2 - \sum_{i=24}^{31} a_{L,i}^2 - \sum_{i=24}^{31} (b_i^{lo})^2 \right)}_{\text{DP8 (4x4)}}
\end{aligned}$$

Step 7 — All components

$$\begin{aligned}
\text{PE} &= 2^8 \sum_{i=0}^7 (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (a_{L,i} + b_i^{hi})^2 + 2^8 \sum_{i=8}^{15} (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (a_{L,i} + b_i^{hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (a_{H,i} + b_i^{lo})^2 + \sum_{i=0}^7 (a_{L,i} + b_i^{lo})^2 + 2^4 \sum_{i=8}^{15} (a_{H,i} + b_i^{lo})^2 + \sum_{i=8}^{15} (a_{L,i} + b_i^{lo})^2 \\
&+ 2^8 \sum_{i=16}^{23} (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=16}^{23} (a_{L,i} + b_i^{hi})^2 + 2^8 \sum_{i=24}^{31} (a_{H,i} + b_i^{hi})^2 + 2^4 \sum_{i=24}^{31} (a_{L,i} + b_i^{hi})^2 \\
&+ 2^4 \sum_{i=16}^{23} (a_{H,i} + b_i^{lo})^2 + \sum_{i=16}^{23} (a_{L,i} + b_i^{lo})^2 + 2^4 \sum_{i=24}^{31} (a_{H,i} + b_i^{lo})^2 + \sum_{i=24}^{31} (a_{L,i} + b_i^{lo})^2 \\
\alpha &= 2^8 \sum_{i=0}^7 a_{H,i}^2 + 2^4 \sum_{i=0}^7 a_{L,i}^2 + 2^8 \sum_{i=8}^{15} a_{H,i}^2 + 2^4 \sum_{i=8}^{15} a_{L,i}^2 \\
&+ 2^4 \sum_{i=0}^7 a_{H,i}^2 + \sum_{i=0}^7 a_{L,i}^2 + 2^4 \sum_{i=8}^{15} a_{H,i}^2 + \sum_{i=8}^{15} a_{L,i}^2 \\
&+ 2^8 \sum_{i=16}^{23} a_{H,i}^2 + 2^4 \sum_{i=16}^{23} a_{L,i}^2 + 2^8 \sum_{i=24}^{31} a_{H,i}^2 + 2^4 \sum_{i=24}^{31} a_{L,i}^2 \\
&+ 2^4 \sum_{i=16}^{23} a_{H,i}^2 + \sum_{i=16}^{23} a_{L,i}^2 + 2^4 \sum_{i=24}^{31} a_{H,i}^2 + \sum_{i=24}^{31} a_{L,i}^2 \\
\beta &= 2^8 \sum_{i=0}^7 (b_i^{hi})^2 + 2^4 \sum_{i=0}^7 (b_i^{hi})^2 + 2^8 \sum_{i=8}^{15} (b_i^{hi})^2 + 2^4 \sum_{i=8}^{15} (b_i^{hi})^2 \\
&+ 2^4 \sum_{i=0}^7 (b_i^{lo})^2 + \sum_{i=0}^7 (b_i^{lo})^2 + 2^4 \sum_{i=8}^{15} (b_i^{lo})^2 + \sum_{i=8}^{15} (b_i^{lo})^2 \\
&+ 2^8 \sum_{i=16}^{23} (b_i^{hi})^2 + 2^4 \sum_{i=16}^{23} (b_i^{hi})^2 + 2^8 \sum_{i=24}^{31} (b_i^{hi})^2 + 2^4 \sum_{i=24}^{31} (b_i^{hi})^2 \\
&+ 2^4 \sum_{i=16}^{23} (b_i^{lo})^2 + \sum_{i=16}^{23} (b_i^{lo})^2 + 2^4 \sum_{i=24}^{31} (b_i^{lo})^2 + \sum_{i=24}^{31} (b_i^{lo})^2
\end{aligned}$$